



Module 14: Database Administration

CPE 011 Database Management System

Slides based on “Database Systems: Design, Implementation, and Management 10th Edition” Chapter 15: Database Administration and Security Slides by Rob, P. and Coronel, C.

Data as a Corporate Asset

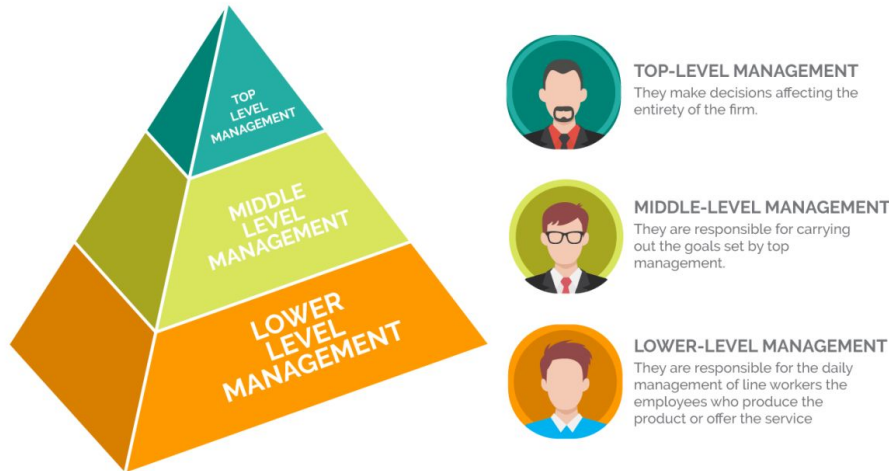
- Data:
 - Valuable asset that requires careful management
 - Valuable resource that translates into information
- **Accurate, timely information** triggers actions that enhance company's position and generate wealth
- Dirty data
 - Data that suffer from inaccuracies and inconsistencies
 - Threat to organizations

Data as a Corporate Asset

- Data quality
 - Comprehensive approach to ensuring the accuracy, validity, and timeliness of the data
- Data profiling software
 - Consists of programs that gather statistics and analyze existing data sources
- Master data management (MDM) software
 - Helps prevent dirty data by coordinating common data across multiple systems.

The Need for and Role of Databases in an Organization

Database's predominant role is to support managerial decision making at all levels.



DBMS facilitates:

- Interpretation and presentation of data
- Distribution of data and information
- Preservation and monitoring of data
- Control over data duplication and use

Three levels to organization management:

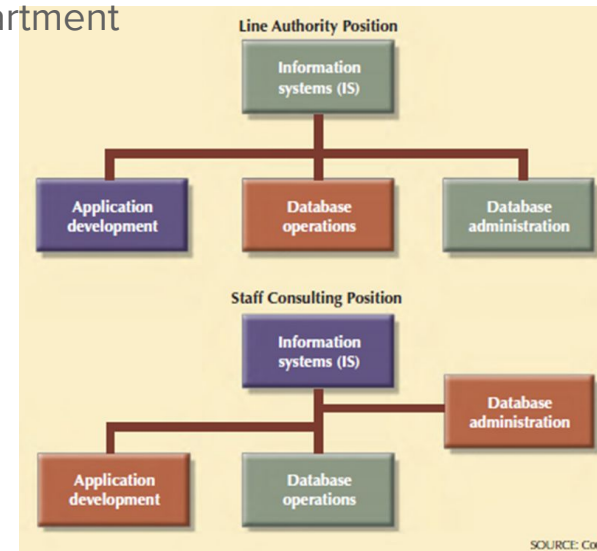
- Top, middle, operational

Database: Special Considerations

- Introduction of a DBMS is likely to have a profound impact (can be positive or negative).
- Three aspects to DBMS introduction:
 - Technological
 - Managerial
 - Cultural
- One role of DBA department is to educate end users about system uses and benefits

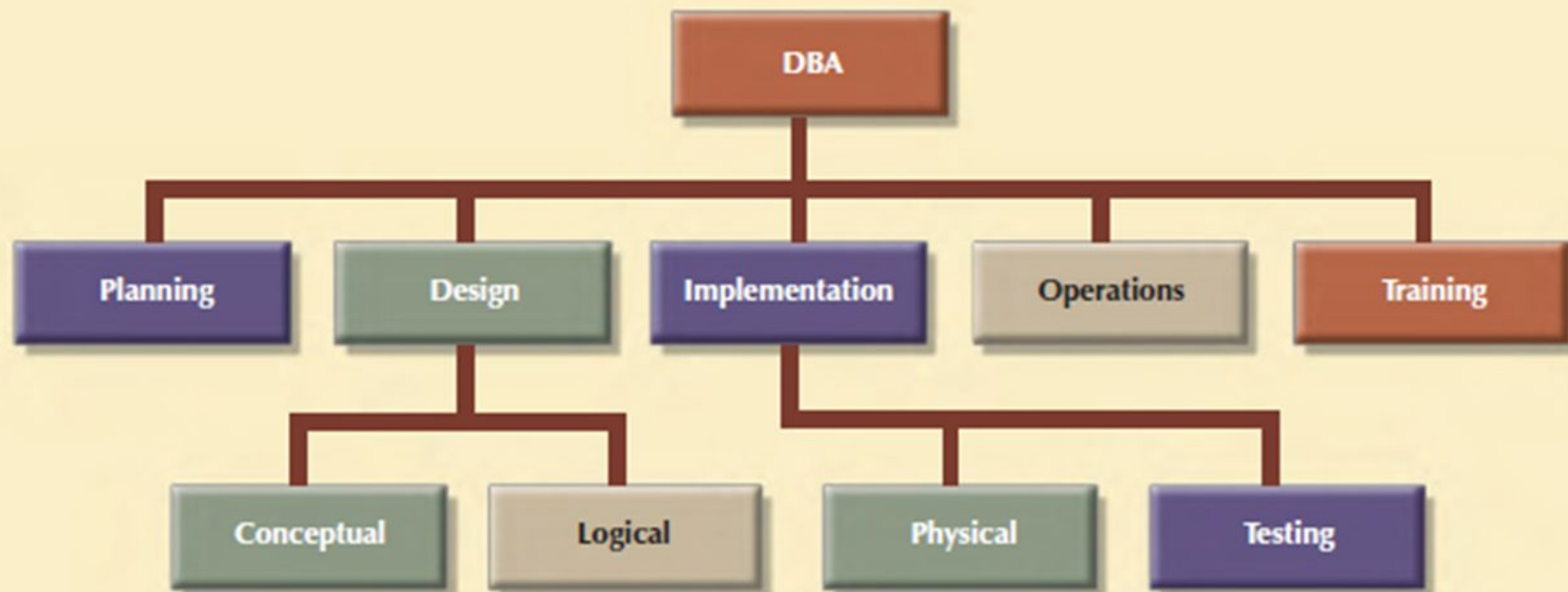
Evolution of Database Administration Function

- Data administration has its roots in the old, decentralized world of the file system
- Advent of DBMS produced new level of data management sophistication
 - DP department evolved into information systems (IS) department
- Data management became increasingly complex
 - Development of database administrator (DBA) function



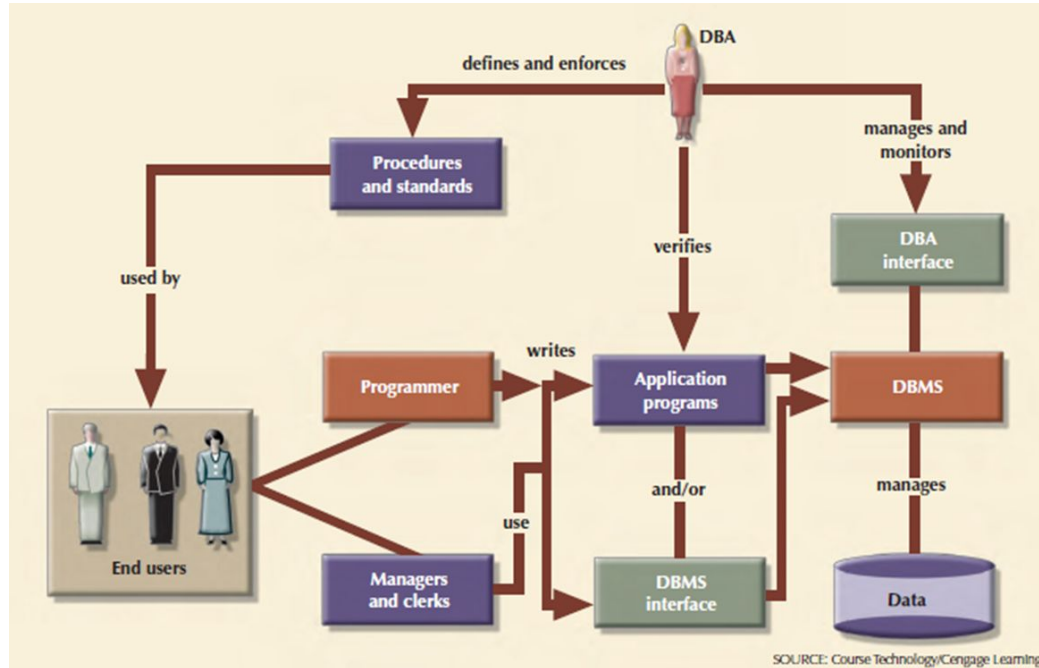
**FIGURE
15.4**

A DBA functional organization



SOURCE: Course Technology/Cengage Learning

Database Environment's Human Component



Even most carefully crafted database system cannot operate without human component.

- Effective data administration requires both technical and managerial skills.
- DA must set data administration goals.
- DBA is focal point for data/user interaction.
- Need for diverse mix of skills.

The DBA's Managerial Role

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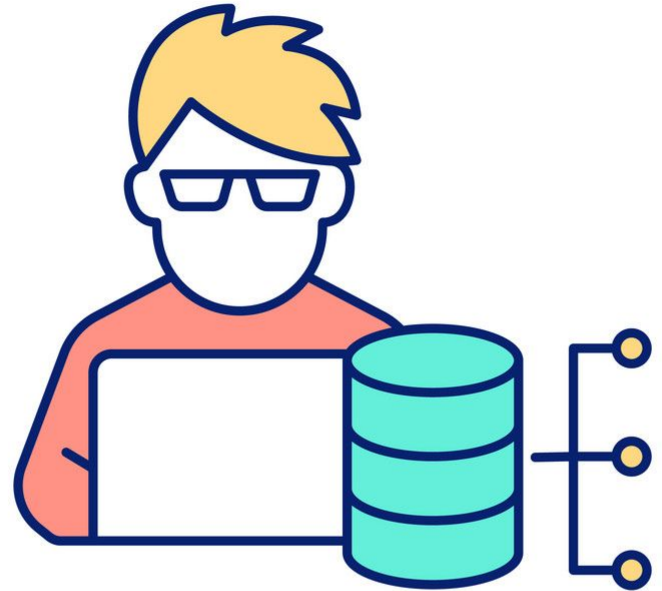
DBA Activities and Services

DBA ACTIVITY	DBA SERVICE
Planning	End-user support
Organizing	Policies, procedures, and standards
Testing	Data security, privacy, and integrity
Monitoring	Data backup and recovery
Delivering	Data distribution and use

- Manages security, privacy, integrity
- Ensures data can be fully recovered (In large organizations, database security officer (DSO) responsible for disaster management)
- Ensures data is distributed appropriately (Makes it easy for authorized end users to access the database)

DBA's Technical Role

- Evaluates, selects, and installs DBMS and related utilities
- Designs and implements databases and applications
- Tests and evaluates databases and applications
- Operates DBMS, utilities, and applications
- Trains and supports users
- Maintains DBMS, utilities, and applications



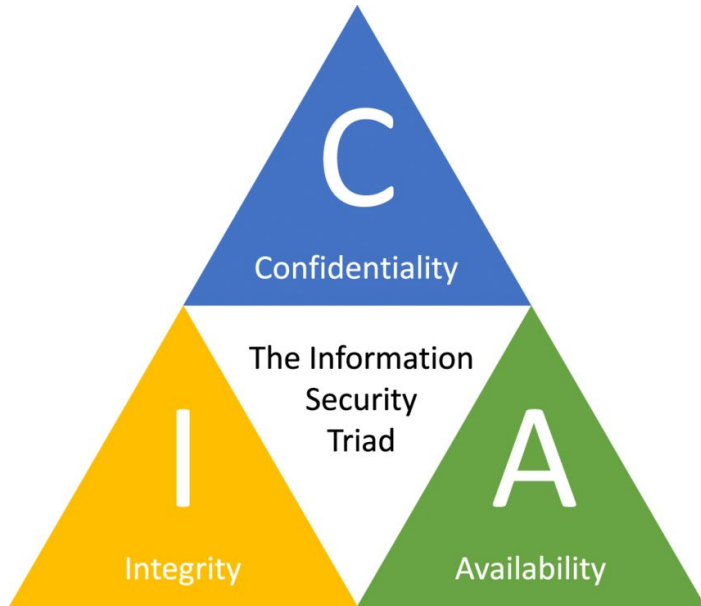
DBA's Role in the Cloud

Cloud services provide:

- DBMS installation and updates
- Server/network management
- Backup and recovery operations

DBA's managerial role is largely unchanged.

Security



Securing data entails securing overall information system architecture.

- **Confidentiality:** data protected against unauthorized access
- **Integrity:** keep data consistent and free of errors or anomalies
- **Availability:** accessibility of data by authorized users for authorized purposes

Security Policies

- Database security officer secures the system and the data
 - Works with the database administrator
- Security policy: collection of standards, policies, procedures to guarantee security
 - Ensures auditing and compliance
- Security audit process identifies security vulnerabilities
 - Identifies measures to protect the system

Security Vulnerabilities

- Security vulnerability: weakness in a system component
 - Could allow unauthorized access or cause service disruptions
- Security threat: imminent security violation
 - Could occur at any time
- Security breach yields a database whose integrity is either:
 - Preserved
 - Corrupted

TABLE
15.4

Sample Security Vulnerabilities and Related Protective Measures

SYSTEM COMPONENT	SECURITY VULNERABILITY	SECURITY MEASURES
People	• The user sets a blank password. • The password is short or includes a birth date. • The user leaves the office door open all the time. • The user leaves payroll information on the screen for long periods of time.	• Enforce complex password policies. • Use multilevel authentication. • Use security screens and screen savers. • Educate users about sensitive data. • Install security cameras. • Use automatic door locks.
Workstation and servers	• The user copies data to a flash drive. • The workstation is used by multiple users. • A power failure crashes the computer. • Unauthorized personnel can use the computer. • Sensitive data are stored on a laptop computer. • Data are lost due to a stolen hard disk or laptop. • A natural disaster occurs.	• Use group policies to restrict the use of flash drives. • Assign user access rights to workstations. • Install uninterrupted power supplies (UPSs). • Add security locks to computers. • Implement a kill switch for stolen laptops. • Create and test data backup and recovery plans. • Insure the system against natural disasters—use co-location strategies.
Operating system	• Buffer overflow attacks • Virus attacks • Root kits and worm attacks • Denial-of-service attacks • Trojan horses • Spyware applications • Password crackers	• Apply OS security patches and updates. • Apply application server patches. • Install antivirus and antispyware software. • Enforce audit trails on the computers. • Perform periodic system backups. • Install only authorized applications. • Use group policies to prevent unauthorized installations.

Applications	• Application bugs—buffer overflow • SQL injection, session hijacking, etc. • Application vulnerabilities—cross-site scripting, nonvalidated inputs • E-mail attacks—spamming, phishing, etc. • Social engineering e-mails	• Test application programs extensively. • Build safeguards into code. • Do extensive vulnerability testing in applications. • Install spam filters and antivirus software for e-mail systems. • Use secure coding techniques (see www.owasp.org). • Educate users about social engineering attacks.
Network	• IP spoofing • Packet sniffers • Hacker attacks • Clear passwords on network	• Install firewalls. • Use virtual private networks (VPNs). • Use intrusion detection systems (IDSs). • Use network access control (NAC). • Use network activity monitoring.
Data	• Data shares are open to all users. • Data can be accessed remotely. • Data can be deleted from a shared resource.	• Implement file system security. • Implement share access security. • Use access permission. • Encrypt data at the file system or database level.

Database Security

- Refers to the use of DBMS features and other measures to comply with security requirements
- DBA secures DBMS from installation through operation and maintenance
- Authorization management
 - User access management
 - View definition
 - DBMS access control
 - DBMS usage monitoring

Thank you for listening!
